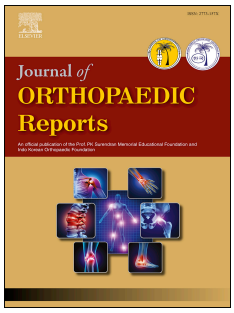
Journal Pre-proof



The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: a quasi-experimental study

Christos Pippas, Nikolaos Dagloukiaris, Antonios Antypas, Emmanouil Touloupis, George Gioftsos

PII: S2773-157X(24)00219-4

DOI: https://doi.org/10.1016/j.jorep.2024.100524

Reference: JOREP 100524

To appear in: *Journal of Orthopaedic Reports*

Received Date: 17 September 2024

Accepted Date: 10 November 2024

Please cite this article as: Pippas C, Dagloukiaris N, Antypas A, Touloupis E, Gioftsos G, The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: a quasi-experimental study, *Journal of Orthopaedic Reports*, https://doi.org/10.1016/j.jorep.2024.100524.

This is a PDF file of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability, but it is not yet the definitive version of record. This version will undergo additional copyediting, typesetting and review before it is published in its final form, but we are providing this version to give early visibility of the article. Please note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2024 Published by Elsevier B.V. on behalf of Prof. PK Surendran Memorial Education Foundation.

Christos Pippas: Conceptualization, Methodology, Formal analysis, Writing – Review & Editing, Supervision, Project administration **Nikolaos Dagloukiaris:** Methodology, Writing – Original draft **Antonios Antypas:** Methodology, Writing – Original draft **Emmanouil Touloupis:** Methodology **George Gioftsos:** Conceptualization, Methodology, Formal analysis, Writing – Review & Editing, Supervision, Project administration

Journal Pre-proof

Journal Pre-proof

The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: a quasi-experimental study

Christos Pippas[a,b,*], Nikolaos Dagloukiaris[a,b], Antonios Antypas[a], Emmanouil Touloupis[b], George Gioftsos[a]

[a] *University of West Attica, Physiotherapy Department, 12243, Athens, Greece*

[b] *Olympiacos FC, 18233, Athens, Greece*

* Corresponding author. University of West Attica, Physiotherapy Department, 28 Agiou Spiridonos str., 12243, Egaleo, Athens, Greece

E-mail addresses: cpippas@uniwa.gr (C. Pippas), ndagloukiaris@gmail.com (N. Dagloukiaris), antonesantypas@gmail.com (A. Antypas), etouloupis1@gmail.com (E. Touloupis), gioftsos@uniwa.gr (G. Gioftsos).

Journal Pre-proof

Journal Pre-proof

The effect of the Copenhagen Adduction and Nordic Hamstring exercises on muscle strength and athletic performance: a quasi-experimental study

Abstract

Background: Two of the most common exercises used in prevention as well as rehabilitation protocols are the Nordic Hamstring (NH) and Copenhagen Adduction (CA). Although their strength effect has been investigated, the effect of the combination of both exercises on muscle strength, jump and sprint performance of male football (soccer) players has yet to be determined.

Materials and Methods: A total of 38 players (15.3±0.49 years, 59.7±7.14 kg, 152.1±53.3 cm) of a Greek football club participated in a progressive 4-week strengthening program, performing both the NH and CA exercises twice a week during the team's pre-season period. Isometric hamstring and eccentric hip adduction strength were assessed using a hand-held dynamometer, while jump and sprint performance were also tested. Perceived exertion was recorded throughout the intervention.

Results: Isometric hamstring strength increased by 11% ($p<0.000$) and eccentric adduction strength increased by 17% ($p<0.000$). There was a decrease of 5-7% ($p=0.000$ to $p=0.034$) in jump performance, and an improvement of 2-3% ($p=0.009$ to $p=0.253$) in sprint performance. There was no statistically significant difference for perceived exertion between the two exercises ($p=0.084$-1.000).

Conclusion: A 4-week progressive exercise program with the Nordic Hamstring and Copenhagen Adduction exercises combined resulted in statistically significant improvements in both isometric hamstring and eccentric adduction strength, but it did not lead to significant results in jump and sprint performance.

Keywords: Copenhagen adduction, Nordic hamstring, strength training, performance

1. Introduction

Football is considered one of the most popular sports in the world[1] with approximately 265 million athletes participating worldwide[2]. The two most frequent muscle injuries in football are those affecting the hamstring and hip adductor muscles[3-5]. Previous investigations have shown that football players who have a previous injury or reduced adductor or hamstring strength are more prone to re-injury[6-10]. Exercise has been shown to be one of the treatments used in the rehabilitation of hamstring and adductor injuries[11, 12]. Two of the exercises commonly used in the sports field and which can improve hamstring[13-16] and adductor muscle strength[17-20] and prevent[21-25], are the Nordic Hamstrings (NH) and Copenhagen Adduction (CA). The NH and CA are two dynamic high-intensity exercises[26, 27] that can be easily implemented on pitch with no equipment needed.

The NH has a controversial effect on sprint and jump capacity. Moran et al.[28] evaluated the effect of the exercise in two programs with different frequencies (once or twice per week) on speed (10m and 40m) and horizontal jump in young football athletes. The results were statistically significant for sprint (10m, $p<0.001$ and 40m, $p=0.001$) but not for jump performance ($p=0.12$). Abdelkader et al.[29] examined the effect of the NH exercise on jump and speed (30m) in 31 pre-adolescent soccer athletes. The results showed improvements in both speed ($p=0.077$) and jumping capacity ($p<0.001$). Ishoi et al.[30] investigated the effect of the same exercise on sprint and eccentric strength of the hamstring muscles in 25 amateur male soccer players. They found a statistically significant improvement of 2.6% ($p<0.05$) in 10m sprint and 19% ($p<0.05$) in eccentric strength. Similarly, Clark et al.[31] found a 6.6% increase ($p=0.04$) in jumping capacity in 9 amateur male football players after four weeks of a NH protocol. Amundsen et al.[14] allocated 32 female soccer players into two groups (high and low load) and evaluated the effect of the NH exercise on hamstring eccentric strength, speed (10m, 20m, 30m, 40m), and jump performance. Both groups showed an increase in eccentric strength

(p=0.001), while neither group demonstrated a statistically significant change in speed (high intensity, p=0.31-0.85; low intensity, p=0.46-0.71) nor jumping (high intensity, p=0.08; low intensity, p=0.69). Siddle et al.[32] examined the effect of a low-load NH program on hamstrings eccentric strength and speed in 16 high-level adolescent soccer players. No improvements in either strength (p≥0.19) or speed (p=0.29-0.47) were observed. In a study by Porrati-Paladino and Cuesta-Barriuso[33], the effect of NH exercise on jumping ability in 15 female soccer athletes was investigated. However, no statistically significant change in the athletes' jumping capacity was observed (p=0.96). Suarez-Arrones et al.[34] examined the effect of the exercise on sprint and hamstring eccentric strength in 50 football athletes. They observed statistically significant improvements of 14.7% in strength (p<0.01) and 4.8% (p<0.01) and 2.9% (p<0.01) in 10m and 20m sprint respectively. Vianna et al.[35] assessed the effect of the NH exercise on eccentric hamstring strength in 16 female soccer players. Both legs showed statistically significant improvements of 13.5% (right: p=0.003; left: p=0.001) after an 8-week training protocol.

The effect of the CA exercise on adductor muscle strength was investigated by Dawkins et al.[17] They used a low-load CA protocol (10-30 repetitions per week for each side) in 39 male football players, with the experimental group showing a 6.7% eccentric strength increase (p=0.03) and a 12.3% isometric strength increase (p<0.01). Kohavi et al.[19] investigated the effect of the CA and Sliding Hip (SH) exercise on eccentric adductor strength in 42 male football players who were allocated into three groups (CA, SH, control). The CA group led to significant increase in eccentric strength for both the right and left leg, with a 45.8% (p<0.001) and 49.4% (p<0.001) improvement respectively. Ishøi et al.[20] investigated the effect of CA exercise on eccentric adductor strength in 20 soccer players, 17 and 18-year-old. The experimental group, which performed the CA twice a week, demonstrated a 35.7% increase (p< 0.001) after eight weeks. Harøy et al.[36] assessed the effect of the CA in combination with the FIFA 11+ program on eccentric hip adductor strength, and the traditional FIFA 11+ program on eccentric hamstring

strength in 33 soccer players. There was an 8% increase ($p<0.001$) in eccentric adductor strength and a 3% increase ($p=0.07$) in eccentric hamstring strength in the CA group, while there was a 7.7% increase ($p<0.001$) in eccentric hamstring strength in the traditional group. No change in speed was observed in either group ($p=0.2$-0.88).

Although the CA and NH exercises play an important role in increasing adductor[17-20] and hamstring muscle strength[13-16] respectively, the effect of the NH on athletic performance remains unclear[14, 28-33] while the effect of the CA on athletic performance is not adequately investigated. To our knowledge, there is no study evaluating the effect of a combination of the NH and CA exercise on muscle strength and athletic performance in young football players.

2. Methods

2.1. Inclusion

2.1.1. Participants

An amateur Greek football club was contacted for participation in the study via e-mail. Healthy male football players 15-16 years old who were systematically (at least four times per week) participating in training and matches were included.

The exclusion criteria were:

(1) the presence of a musculoskeletal injury and/or a neurological disorder that would affect strength and/or sprint/jump testing, and/or the intervention.

(2) an injury of the hamstrings and/or the adductor muscles 6 months before study initiation.

(3) pain greater than 2/10 on a Numeric Rating Scale (NRS) during strength and/or athletic performance testing, and/or during the intervention.

(4) any systematic performance of hamstring and/or adductor resistance exercises during

the last month prior to study initiation.

2.1.2. Ethics

Approval was obtained from the University of West Attica ethics committee (77055/01-09-2023) and written informed consent was obtained for every participant in accordance with the Declaration of Helsinki.

2.1.3. Blinding

Participants as well as exercise supervisors were blinded to strength, sprint, and jump testing results.

2.2. Intervention

The intervention duration was four weeks and was performed during the team's pre-season period in the club's facilities. Participants performed a progressive protocol consisting of a combination of the NH and CA exercises, twice a week, with a break of at least 24 hours between sessions. The intervention was conducted as part of the team's condition training before the main training program. To ensure proper execution of the exercises, a physiotherapist and a fitness coach supervised the intervention.

Nordic Hamstring exercise

The NH has been previously described[37]. It is performed in pairs with athletes of similar height and weight. The athlete performing the exercise is placed on their knees stabilized by a teammate (Figure 1). The athlete crosses their hands in front of their chest with their elbows bent and palms facing their shoulders. The athlete starts by slowly and steadily lowering their body to the ground while keeping their trunk aligned with their thighs. Once the athlete reaches

Journal Pre-proof

the limit where they cannot longer resist the fall, they use their arms to push themselves back to the start position.

Figure 1. The Nordic Hamstring exercise. (A) Start position, (B) End position.

Copenhagen Adduction exercise

The CA has been previously described[27]. It is performed in pairs with athletes of similar height and weight. The athlete lies on their side with the forearm of the supporting side on the ground and the other arm placed on the pelvis (Figure 2). The teammate holds the training leg at the level of the pelvis holding under the knee and ankle. The player brings their pelvis up in a 3-second concentric movement until it reaches a straight line, bringing the lower leg in adduction until it touches the training leg. Then they lower the pelvis towards the ground in a controlled manner and bring the lower leg to abduction in a 3-second eccentric movement until it touches the ground without supporting the body. Any flexion, side flexion, and rotation of the trunk and/or hips should be avoided. The exercise should be performed in both sides.

Figure 2. The Copenhagen Adduction exercise. (A) Start and end position, (B) Mid position.

2.2.1. Training Load

Moran et al.[28] and Váczi et al.[38] performed a NH training with 2 sets of 5 repetitions for the first week. Moran et al.[28] increased the repetitions up to 8 in week 7, while Váczi et al.[38] progressively increased the repetitions up to 15 in weeks 9 and 10. Both studies defined a set range of 2 to 3 throughout the intervention. Alonso-Calvete et al.[39] maintained 2 sets of the CA exercise throughout an 8-week intervention and progressively increased repetitions (6-15). Ishoi and Thorborg[40] suggested a linear relationship between eccentric adduction strength and the CA training volume. In addition, we tried to achieve similar training volume (total repetition

number) between the two exercises (Table 1). Detailed description of the exercise characteristics is provided in Appendix A.

Table 1. Copenhagen Adduction and Nordic hamstring exercise training protocol

2.2.2. Outcome measures

The primary outcome measures were eccentric hip adductor muscle strength (EHAD) and isometric hamstring strength (IH). The secondary outcome measures were jump performance (SJ, CMJ, and CMJh), sprint performance (5m, 10m, 20m), and perceived exertion. The K-Force Muscle Controller hand-held dynamometer (KINVENT, Montpellier, France) was used to measure muscle strength based on reliable procedures[10, 41-44]. Strength and performance testing was conducted by people not otherwise involved in the study. Jump and sprint performance were assessed with the use of the Microgate Optogait (Microgate, Bolzano, Italy) and Fitlight System (Fitlight USA Inc., Miami, Florida) respectively. Both strength and performance testing was conducted before and after the 4-week intervention period in the club's facilities. The athletes refrained from any form of training 24 hours prior to the measurements. The description of strength as well as performance testing is provided in Appendix B.

2.2.3. Compliance

Compliance was recorded at the beginning of each exercise session.

2.2.4. Perceived Exertion

Perceived exertion was recorded at the end of each exercise session with the use of the Borg scale (0-10)[45], with the players reporting exertion scores for each exercise separately.

2.3. Statistical analysis

Data are presented as mean ± standard deviation (mean ± SD). Both intention-to-treat (ITT) and per-protocol (PP) analyses were used. Athletes who attended less than 75% of the total sessions were excluded for the PP analysis. Multiple imputation was used for missing data in the ITT analysis. A paired samples t-test was used to determine the effect of time (baseline vs follow-up) on the dependent variables (strength, jump, and sprint performance). Repeated measures ANOVA with Bonferroni correction was used to determine the difference between exercises for perceived exertion. The effect size (Cohen's d) was classified as small (0.2), medium (0.5), or large (0.8)[46]. Data analysis was conducted using SPSS Statistics (v 29, IBM Corporation).

2.4. Sample size

An a priori sample size calculation was performed with the use of G*power software (v. 3.1, Heinrich-Heine-Universität, Düsseldorf, Germany). In line with previous studies, the minimal important difference (MID) for eccentric adduction strength was set at 15%, while a 25% difference was set for isometric hamstring strength. With a significance level (α) of 0.5, statistical power of 80%, and an effect size of 0.5, a sample size of at least 27 participants was required. To account for dropouts we attempted to include 35 subjects.

3. Results

3.1. Participants

One Greek football club was included in the study. The inclusion process is presented in Figure 3. Overall, 38 players were included for analysis. Participants demographics are presented in Table 2. Strength testing and athletic performance testing were conducted in different days due to time limitations. 4 players lost both baseline and follow-up performance testing and therefore

38 players performed strength testing and 34 players performed jump and sprint testing in the ITT analysis. Follow-up performance testing was lost for 12 players and strength testing for 4 players for reasons not related to the intervention. One player lost baseline performance testing. Compliance was less than 75% for 10 players, 5 of which also had missing follow-up performance measurements. All 10 players were excluded from the analysis. In the PP analysis 24 players performed strength testing and 16 players performed jump and sprint testing.

Figure 3. Flowchart of participants inclusion process

Table 2. Participants demographics

3.2. Strength effects

In the ITT analysis, there was a 17% difference in EHAD between baseline and follow-up measurements (from 182.7±48.3 N to 214.6±47.7 N, $p<0.000$) (Table 3, Figure 4). In the PP analysis, there was a 20% difference in EHAD between baseline and follow-up measurements (from 177.5±44.8 N to 213.7±46.5 N, $p<0.000$) (Table 3, Figure 4).

In the ITT analysis, there was an 11% difference in IH between baseline and follow-up measurements (from 289.4±49.2 N to 321±60.5 N, $p<0.000$) (Table 3, Figure 5). In the PP analysis, there was a 10% difference in IH between baseline and follow-up measurements (from 288.2±56.5 N to 314.7±65 N, $p=0.001$) (Table 3, Figure 5).

Journal Pre-proof

Journal Pre-proof

220 *Table 3*. Exercise effects on adductor and hamstring strength

Journal Pre-proof

Figure 4. Eccentric hip adduction strength.

Figure 5. Isometric hamstring strength.

3.3. Jump and sprint capacity

In the ITT analysis, there was a decrease in SJ jump from 27.34±3.8 to 24.37±3.5 cm (p<0.000), corresponding to a 10% difference. Similarly, a decrease of 5% was observed in the CMJ jump, from 27.50±3.8 to 25.97±3.5 cm (p<0.034). The CMJh jump also decreased by 5%, from 32.89±3.9 to 31.02±3.6 cm (p=0.013). Jump results are presented in Appendix C.

In the ITT analysis, there was no significant improvement in 5m sprint, with a decrease from 1.10±0.06 s to 1.08±0.05 s (p=0.253), a difference of 2%. However, there was a statistically significant decrease in 10m sprint from 1.92±0.10 s to 1.87±0.07 s (p=0.009), a difference of 3%. Additionally, a non-significant decrease in time of 2% was observed for the 20m sprint from 3.36±0.19 s to 3.30±0.15 s (p=0.072). Sprint results are presented in Appendix D.

3.4. Compliance

There was a compliance of 90% for the ITT analysis and a compliance of 77% for the PP analysis.

3.5. Perceived exertion

In the ITT analysis there was no statistically significant difference between the two exercises (p=0.084-1.000). The median value for the CA exercise showed a range of 3.5-4 and for the NH exercise a range of 3-4. In PP analysis there was also no statistically significant difference

between the two exercises (p=0.278-1.000). The median value for the CA exercise showed a range of 3.5-4 and for the NH exercise a range of 3-4. The results for perceived exertion are presented in Appendix E.

4. Discussion

The current study assessed the effect of the combination of the Copenhagen Adduction and Nordic Hamstring exercise on muscle torque and athletic performance in young soccer players. The results showed statistically significant increases both for adductor and hamstring torque. No significant improvements were observed for jump and sprint performance.

Strength effects

The findings of the current study showed a statistically significant EHAD increase of 17% (d=0.66). This increase is consistent with the findings of Dawkins at el.[17] who observed a 7% increase in adductor muscle strength after a 6-week CA protocol. Ishoi and Thorborg[40] indicated the linearity between training volume and adduction strength gains. In the current study the total repetition number of the CA was 288, while Dawkins at el.[17] applied a total volume of 220 repetitions. Harøy et al.[36] observed an 8% increase in eccentric adduction strength after 8 weeks of the CA with a total number of 480 repetitions, while Ishøi et al.[20] found a 36% increase with a training volume of 960 repetitions. The comparison of the strength increase of the current study with this of the aforementioned studies indicates the possibility that the difference in strength gains might be the result of the combination of the CA and the NH exercise. It has been previously hypothesized that there is a relationship between the adductors and the hamstring muscles[47, 48]. In addition, it has been shown that the adductor longus is activated during hip extension[49, 50]. We could therefore assume the presence of co-activation of the adductor and the hamstrings during hamstring exercises and the transfer of the

adductor strength gains to these of the hamstring muscles and vice versa.

The current study found an 11% increase in isometric hamstring strength. Previous investigations evaluated the effect of the NH exercise on hamstring eccentric[13, 14, 30], reporting an increase ranging from 4-36%. Timmins et al.[51] evaluated the effect of the NH on isometric strength and found a statistically significant decrease of 13% (p=0.02).

Jump capacity

The current study showed a decrease of 5-10% in jump performance, which supports the findings of previous studies. Amundsen et al.[14] conducted an 8-week study including two groups of the NH exercise: a high-volume training group and a low-volume training group. The high-volume group noted a 2.5% decrease in jump performance (p=0.08), while the low volume group noted a non-statistically significant increase of less than 1% (p=0.09). Similarly, Porrati-Paladino and Cuesta-Barriuso[33] observed a non-statistically significant improvement in jumping of less than 1% with the use of the NH exercise over a 6-week training period. In contrast, Moran et al.[28] used an 8-week NH protocol and found an improvement of 12% (p=0.12). Abdelkader et al.[29] included two exercise groups of different frequency (one and two times per week) for 8 weeks. The groups demonstrated a statistically significant improvement of 7% (p=0.003) and 13% (p<0.001) respectively.

Sprint capacity

The current study showed an improvement of 2%, 3%, and 2% for 5m, 10m, and 20m, respectively. However, the improvement was not statistically significant for the 5m and 20m sprint, whereas it was for the 10m sprint. Our results are consistent with those of Amundsen et al.[14] who found a non-statistically significant increase in time of less than 1% at 10m, 20m,

30m, and 40m (p=0.31-0.85). Similarly, Siddle et al.[32] used an 8-week NH program and found an increase in time of less than 1% (p=0.29-0.47), while Harøy et al.[36] also found non-statistically significant differences of less than 1% (p=0.2-0.88) using the FIFA 11+ program for 8 weeks. In contrast, Abdelkader et al.[29] observed a 3% time decrease in the experimental group (p=0.077) and a 6% time increase in the control group (p=0.031) in the 30m sprint after an 8-week exercise protocol. In addition, Moran et al.[28] found an improvement of 7.6% (p<0.001) and 2.3% (p<0.001) at the 10m and 40m sprint respectively.

Perceived exertion

No between-group differences were observed for neither analysis, which might support the hypothesis that the similar training volume of the exercises led to similar exertion levels. The mean exertion value for the CA was 3.9, with a range of 0-9. The highest value was observed in the 7th and 8th sessions, where the players performed 2 sets of 12 repetitions of the exercise. Dawkins et al.[17] reported a mean range of 3-7, with an individual range of 0-9, and the highest value recorded in the 9th session. Exercise dosage at that time was 1 set of 12 repetitions. Polglass, Burrows, and Willett[18] reported a mean exertion value of 2.6, with an individual range of 1-6 and the highest value noted in the 9th session, where the players performed 3 sets of 8 repetitions. Ishoi et al.[20] observed a mean exertion value of 5, with an individual range of 1 and 10. The highest value was noted in the 9th and 13th session with 3 sets of 12 and 15 repetitions respectively. For the NH exercise, the mean value was 3.5, with an individual range of 0-8, and the highest value was noted in the 2nd, 5th, 7th, and 8th session, where the players performed 2 sets of 5 repetitions, 3 sets of 5 repetitions, and 3 sets of 7 repetitions. However, there is no study evaluating exertion levels during the use of a NH exercise protocol.

Study limitations

A major limitation of the current study is its design which did not enable randomization to be performed as only one group was included. As in similar study designs, internal validity might have been compromised and therefore the results should be used with caution. In addition, the absence of a control group weakens even further the examination of the causal relationship of the study. Another limitation is that the baseline and performance measurements were conducted in different testing occasions, which poses a reason for the high rate of the players who lost to follow-up and this might have led to type II error.

The use of randomized controlled trials is recommended so that more definite conclusions can be drawn. Furthermore, a better designed methodology should be planed so that high lost to follow-up rates are avoided.

Conclusions

Statistically significant increases in eccentric hip adduction and isometric hamstring strength in young football players performing the Copenhagen Adduction and the Nordic Hamstring exercise were observed. No statistically significant improvements neither in jump nor sprint performance were recorded.

Acknowledgments: The authors would like to thank the club participating in the study as well as each and every individual player for their cooperation. Additionally, the authors would like to thank every person for their contribution to the completion of the study.

Journal Pre-proof

References

1. Werner J, Hagglund M, Walden M, Ekstrand, J. UEFA Injury Study: A Prospective Study of Hip and Groin Injuries in Professional Football over Seven Consecutive Seasons. *Br J Sports Med* 2009, *43* (13), 1036–1040. https://doi.org/10.1136/bjsm.2009.066944.
2. Kunz M. *265 million playing football.* FIFA Magazine
3. Ekstrand J, Hägglund M, Waldén M. Epidemiology of Muscle Injuries in Professional Football (Soccer). *Am J Sports Med* 2011, *39* (6), 1226–1232.
4. HÃ¤gglund M, WaldÃ©n M, Ekstrand J. Injuries among Male and Female Elite Football Players. *Scand J Med Sci Sports* 2009, *19* (6), 819–827. https://doi.org/10.1111/j.1600-0838.2008.00861.x.
5. Nielsen AB, Yde J. Epidemiology and Traumatology of Injuries in Soccer. *Am J Sports Med* 1989, *17* (6), 803–807. https://doi.org/10.1177/036354658901700614.
6. Liu H, Garrett WE, Moorman CT, Yu B. Injury Rate, Mechanism, and Risk Factors of Hamstring Strain Injuries in Sports: A Review of the Literature. *J Sport Health Sci* 2012, *1* (2), 92–101. https://doi.org/10.1016/j.jshs.2012.07.003.
7. Crow JF, Pearce AJ, Veale JP, Vander Westhuizen D, Coburn PT, Pizzari T. Hip Adductor Muscle Strength Is Reduced Preceding and during the Onset of Groin Pain in Elite Junior Australian Football Players. *J Sci Med Sport* 2010, *13* (2), 202–204. https://doi.org/10.1016/j.jsams.2009.03.007.
8. Engebretsen AH, Myklebust G, Holme I, Engebretsen L, Bahr R. Intrinsic Risk Factors for Groin Injuries among Male Soccer Players. *Am J Sports Med* 19892010, *38* (10), 2051–2057. https://doi.org/10.1177/0363546510375544.
9. O'Connor DM. Groin Injuries in Professional Rugby League Players: A Prospective Study. *J Sports Sci* 2004, *22* (7), 629–636.

Journal Pre-proof

https://doi.org/10.1080/02640410310001655804.

10. Tyler TF, Nicholas SJ, Campbell RJ, McHugh MP. The Association of Hip Strength and Flexibility with the Incidence of Adductor Muscle Strains in Professional Ice Hockey Players. *Am J Sports Med* 2001, *29* (2), 124–128. https://doi.org/10.1177/03635465010290020301.
11. Hickey JT, Opar DA, Weiss LJ, Heiderscheit BC. Current Clinical Concepts: Hamstring Strain Injury Rehabilitation. *J Athl Train* 2021, *57* (2). https://doi.org/10.4085/1062-6050-0707.20.
12. Thorborg K, Sportsphysio M. Current Clinical Concepts: Exercise and Load Management of Adductor Strains, Adductor Ruptures and Longstanding Adductor-Related Groin Pain. *J Athl Train* 2022, *58* (7-8). https://doi.org/10.4085/1062-6050-0496.21.
13. Ripley NJ, Cuthbert M, Comfort P, McMahon JJ. Effect of Additional Nordic Hamstring Exercise or Sprint Training on the Modifiable Risk Factors of Hamstring Strain Injuries and Performance. *PLOS ONE* 2023, *18* (3), e0281966. https://doi.org/10.1371/journal.pone.0281966.
14. Amundsen R, Heimland JS, Thorarinsdottir S, Møller M, Bahr R. Effects of High and Low Training Volume with the Nordic Hamstring Exercise on Hamstring Strength, Jump Height, and Sprint Performance in Female Football Players: A Randomised Trial. *Transl Sports Med*2022, *2022*, 1–9. https://doi.org/10.1155/2022/7133928.
15. Marchiori CL, Medeiros DM, Severo-Silveira L, dos Santos Oliveira G, Medeiros TM, de Araujo Ribeiro-Alvares JB, et al. Muscular Adaptations to Training Programs Using the Nordic Hamstring Exercise or the Stiff-Leg Deadlift in Rugby Players. *Sport Sci Health* 2021, *18* (2). https://doi.org/10.1007/s11332-021-00820-0.
16. Alt T, Nodler YT, Severin J, Knicker AJ, Strüder HK. Velocity-Specific and Time-

Journal Pre-proof

Dependent Adaptations Following a Standardized Nordic Hamstring Exercise Training. *Scand J Med Sci Sports*2017, *28* (1), 65–76. https://doi.org/10.1111/sms.12868.

17. Dawkins J, Ishøi L, Willott JO, Andersen LL, Thorborg K. Effects of a Low-Dose Copenhagen Adduction Exercise Intervention on Adduction Strength in Sub-Elite Male Footballers: A Randomised Controlled Trial. *Transl Sports Med*2021, *4* (4). https://doi.org/10.1002/tsm2.238.
18. Polglass G, Burrows A, Willett M. Impact of a Modified Progressive Copenhagen Adduction Exercise Programme on Hip Adduction Strength and Postexercise Muscle Soreness in Professional Footballers. *BMJ Open SEM* 2019, *5* (1), e000570. https://doi.org/10.1136/bmjsem-2019-000570.
19. Kohavi B, Beato M, Laver L, Freitas TT, Chung LH, Dello Iacono A. Effectiveness of Field-Based Resistance Training Protocols on Hip Muscle Strength among Young Elite Football Players. *Clin J Sport Med* 2018, 1. https://doi.org/10.1097/jsm.0000000000000649.
20. Ishøi L, Sørensen CN, Kaae NM, Jørgensen LB, Hölmich P, Serner A. Large Eccentric Strength Increase Using the Copenhagen Adduction Exercise in Football: A Randomized Controlled Trial. *Scand J Med Sci Sports* 2015, *26* (11), 1334–1342. https://doi.org/10.1111/sms.12585.
21. van Dyk N, Behan FP, Whiteley R. Including the Nordic Hamstring Exercise in Injury Prevention Programmes Halves the Rate of Hamstring Injuries: A Systematic Review and Meta-Analysis of 8459 Athletes. *Br J Sports Med* 2019, *53* (21), 1362–1370. https://doi.org/10.1136/bjsports-2018-100045.
22. Harøy J, Clarsen B, Wiger EG, Øyen MG, Serner A, Thorborg K, et al. The Adductor Strengthening Programme Prevents Groin Problems among Male Football Players: A Cluster-Randomised Controlled Trial. *Br J Sports Med* 2018, *53* (3), 150–157.

Journal Pre-proof

https://doi.org/10.1136/bjsports-2017-098937.

23. Ribeiro-Alvares JB, Marques VB, Vaz MA, Baroni BM. Four Weeks of Nordic Hamstring Exercise Reduce Muscle Injury Risk Factors in Young Adults. *J Strength Cond Res* 2018, *32* (5), 1254–1262. https://doi.org/10.1519/jsc.0000000000001975.
24. Bourne MN, Duhig SJ, Timmins RG, Williams MD, Opar DA, Al Najjar A, et al. Impact of the Nordic Hamstring and Hip Extension Exercises on Hamstring Architecture and Morphology: Implications for Injury Prevention. *Br J Sports Med* 2016, *51* (5), 469–477. https://doi.org/10.1136/bjsports-2016-096130.
25. van der Horst N, Smits D-W, Petersen J, Goedhart EA, Backx FJG. The Preventive Effect of the Nordic Hamstring Exercise on Hamstring Injuries in Amateur Soccer Players. *Am J Sports Med* 2015, *43* (6), 1316–1323.
26. Llurda-Almuzara L, Labata-Lezaun N, López-de-Celis C, Aiguadé-Aiguadé R, Romaní-Sánchez S, Rodríguez-Sanz J, et al. Biceps Femoris Activation during Hamstring Strength Exercises: A Systematic Review. *Int J Environ Health Res* 2021, *18* (16), 8733. https://doi.org/10.3390/ijerph18168733.
27. Serner A, Jakobsen MD, Andersen LL, Hölmich P, Sundstrup E, Thorborg K. EMG Evaluation of Hip Adduction Exercises for Soccer Players: Implications for Exercise Selection in Prevention and Treatment of Groin Injuries. *Br J Sports Med* 2013, *48* (14), 1108–1114. https://doi.org/10.1136/bjsports-2012-091746.
28. Moran J, Vali N, Drury B, Hammami R, Tallent J, Chaabene H, et al. The Effect of Volume Equated 1- versus 2-Day Formats of Nordic Hamstring Exercise Training on Fitness in Youth Soccer Players: A Randomised Controlled Trial. *PloS One* 2022, *17* (12), e0277437. https://doi.org/10.1371/journal.pone.0277437.
29. Abdelkader M, Hammami R, Drury B, Clark N, Sandercock G, Shaw I, et al. A Randomised Controlled Trial of 1- versus 2-Day per Week Formats of Nordic

Journal Pre-proof

Hamstring Training on Explosive Athletic Tasks in Prepubertal Soccer Players. *J Sports Sci* 2022, *40* (19), 2173–2181. https://doi.org/10.1080/02640414.2022.2145737.

30. Ishøi L, Hölmich P, Aagaard P, Thorborg K, Bandholm T, Serner A. Effects of the Nordic Hamstring Exercise on Sprint Capacity in Male Football Players: A Randomized Controlled Trial. *J Sports Sci* 2017, *36* (14), 1663–1672. https://doi.org/10.1080/02640414.2017.1409609.
31. Clark R, Bryant A, Culgan J-P, Hartley B. The Effects of Eccentric Hamstring Strength Training on Dynamic Jumping Performance and Isokinetic Strength Parameters: A Pilot Study on the Implications for the Prevention of Hamstring Injuries. *Phys Ther Sport* 2005, *6* (2), 67–73. https://doi.org/10.1016/j.ptsp.2005.02.003.
32. Siddle J, Weaver K, Greig M, Harper D, Brogden CM. A Low-Volume Nordic Hamstring Curl Programme Improves Change of Direction Ability, despite No Architectural, Strength or Speed Adaptations in Elite Youth Soccer Players. *Res Sports Med* 2022, *32* (1), 1–12. https://doi.org/10.1080/15438627.2022.2079984.
33. Porrati-Paladino G, Cuesta-Barriuso R. Effectiveness of Plyometric and Eccentric Exercise for Jumping and Stability in Female Soccer Players—a Single-Blind, Randomized Controlled Pilot Study. *Int J Environ Health Res* 2021, *18* (1), 294. https://doi.org/10.3390/ijerph18010294.
34. Suarez-Arrones L, Lara-Lopez P, Rodriguez-Sanchez P, Lazaro-Ramirez JL, Di Salvo V, Guitart M, et al. Dissociation between Changes in Sprinting Performance and Nordic Hamstring Strength in Professional Male Football Players. *PLOS ONE* 2019, *14* (3), e0213375. https://doi.org/10.1371/journal.pone.0213375.
35. Vianna KB, Rodrigues LG, Oliveira NT, Ribeiro-Alvares JB, Baroni BM. A Preseason Training Program with the Nordic Hamstring Exercise Increases Eccentric Knee Flexor Strength and Fascicle Length in Professional Female Soccer Players. *Int J Sports Phys*

Journal Pre-proof

Ther 2021, *16* (2). https://doi.org/10.26603/001c.19452.

36. Harøy J, Thorborg K, Serner A, Bjørkheim A, Rolstad LE, Hölmich P, et al. Including the Copenhagen Adduction Exercise in the FIFA 11+ Provides Missing Eccentric Hip Adduction Strength Effect in Male Soccer Players: A Randomized Controlled Trial. *Am J Sports Med* 2017, *45* (13), 3052–3059. https://doi.org/10.1177/0363546517720194.
37. Mjølsnes R, Arnason A, Østhagen T, Raastad T, Bahr RA. 10-week Randomized Trial Comparing Eccentric vs. Concentric Hamstring Strength Training in Well-Trained Soccer Players. *Scand J Med Sci Sports* 2004 https://pubmed.ncbi.nlm.nih.gov/15387805/.
38. Váczi M, Fazekas G, Pilissy T, Cselkó A, Trzaskoma L, Sebesi B, et al. The Effects of Eccentric Hamstring Exercise Training in Young Female Handball Players. *Eur J Appl Physiol* 2022, *122* (4). https://doi.org/10.1007/s00421-022-04888-5.
39. Alonso-Calvete A, Lorenzo-Martínez M, Padrón-Cabo A, Rey E. Effects of Copenhagen Adduction Exercise on the Architectural Characteristics of Adductors in U-17 Male Soccer Players: A Randomized Controlled Trial. *Int J Environ Health Res* 2021, *18* (24), 12956. https://doi.org/10.3390/ijerph182412956.
40. Ishøi L, Thorborg K. Copenhagen Adduction Exercise Can Increase Eccentric Strength and Mitigate the Risk of Groin Problems: But How Much Is Enough! *Br J Sports Med* 2021, bjsports-2020-103564. https://doi.org/10.1136/bjsports-2020-103564.
41. Pippas C, Emmanouilidis A, Karanasios S, Koumantakis G, Gioftsos G. Reliability of the K-Force Muscle Controller Dynamometer on Eccentric and Isometric Hip Adduction Strength. *Open J Ther Rehabil* 2024, *12* (3), 224–235. https://doi.org/10.4236/ojtr.2024.123017.
42. Serner A, Weir A, Tol JL, Thorborg K, Lanzinger S, Otten R, et al. Return to Sport after Criteria-Based Rehabilitation of Acute Adductor Injuries in Male Athletes: A

Journal Pre-proof

Prospective Cohort Study. *Orthop J Sports Med* 2020, *8* (1). https://doi.org/10.1177/2325967119897247.

43. Ishøi L, Hölmich P, Thorborg K. MEASURES of HIP MUSCLE STRENGTH and RATE of FORCE DEVELOPMENT USING a FIXATED HANDHELD DYNAMOMETER: INTRA-TESTER INTRA-DAY RELIABILITY of a CLINICAL SET-UP. *Int J Sports Phys Ther* 2019, *14* (5), 715–723.
44. McCall A, Nedelec M, Carling C, Le Gall F, Berthoin S, Dupont G. Reliability and Sensitivity of a Simple Isometric Posterior Lower Limb Muscle Test in Professional Football Players. *J Sports Sci* 2015, *33* (12), 1298–1304. https://doi.org/10.1080/02640414.2015.1022579.
45. Borg E, Borg G, Larsson K, Letzter M, Sundblad B-M. An Index for Breathlessness and Leg Fatigue. *Scand J Med Sci Sports* 2010, *20* (4), 644–650. https://doi.org/10.1111/j.1600-0838.2009.00985.x.
46. Cohen J. A Power Primer. *Psychol Bull* 1992, *112* (1), 155–159. https://doi.org/10.1037//0033-2909.112.1.155.
47. Obey MR, Broski SM, Spinner RJ, Collins MS, Krych AJ. Anatomy of the Adductor Magnus Origin. *Orthop J Sports Med* 2016, *4* (1), 232596711562505. https://doi.org/10.1177/2325967115625055.
48. Broski SM, Murthy NS. Krych AJ, Obey MR, Collins MS. The Adductor Magnus "Mini-Hamstring": MRI Appearance and Potential Pitfalls. *Skeletal Radiol* 2015, *45* (2), 213–219. https://doi.org/10.1007/s00256-015-2291-5.
49. Kato T, Taniguchi K, Akima H, Watanabe K, Ikeda Y, Katayose M. Effect of Hip Angle on Neuromuscular Activation of the Adductor Longus and Adductor Magnus Muscles during Isometric Hip Flexion and Extension. *Eur J Appl Physiol* 2019, *119* (7), 1611–1617. https://doi.org/10.1007/s00421-019-04150-5.

Journal Pre-proof

50. Benn ML, Pizzari T, Rath L, Tucker K, Semciw AI. Adductor Magnus: An EMG Investigation into Proximal and Distal Portions and Direction Specific Action. *Clin Anat* 2018, *31* (4), 535–543. https://doi.org/10.1002/ca.23068.

51. Timmins RG, Filopoulos D, Giannakis J, Nguyen V, Ruddy JD, Hickey JT, et al. The Impact of Eccentric or Isometric Training on Strength, Architecture, and Sprinting across an Australian Football Season. *Med Sci Sports Exerc* 2023. https://doi.org/10.1249/MSS.0000000000003326.

52. França C, Gouveia É, Caldeira R, Marques A, Martins J, Lopes H, et al. Speed and Agility Predictors among Adolescent Male Football Players. *Int J Environ Health Res* 2022, 19 (5), 2856. https://doi.org/10.3390/ijerph19052856.

53. Bogataj Š, Pajek M, Hadžić V, Andrašić S, Padulo J, Trajković N. Validity, Reliability, and Usefulness of My Jump 2 App for Measuring Vertical Jump in Primary School Children. *Int J Environ Health Res* 2020, 17 (10), 3708. https://doi.org/10.3390/ijerph17103708.

Journal Pre-proof

533

Journal Pre-proof

534

Journal Pre-proof

535

Journal Pre-proof

Table 1. Copenhagen Adduction and Nordic hamstring exercise training protocol

Week	Copenhagen Adduction			Nordic Hamstring		
	Session/week	Set/side	Repetitions/side	Session/week	Set	Repetitions
1	2	2	6	2	2	5
2	2	2	8	2	2	7
3	2	2	10	2	3	5
4	2	2	12	2	3	7
Total	**8**	**8**	**144**	**8**	**10**	**120**

Table 2. Participants demographics

	ITT	PP
Age (years)	15.4±0.50 (15-16)	15.3±0.49 (15-16)
Weight (kgs)	60.6±7.89 (41.4-83.1)	59.7±7.14 (41.4-73.4)
Height (cm)	170.8±6.86 (154-190)	170.4±7.30 (154-190)

Presented as mean±SD (range). Kgs; kilograms. Cm; centimeters.

Journal Pre-proof

Journal Pre-proof

Table 3. Exercise effects on adductor and hamstring strength

	Baseline	Follow-up	Mean Difference (95% CI)	Effect Size (% difference)	p value
Intention-to-Treat					
Eccentric adduction strength	182.7 (±48.3)	214.6 (±47.7)	31.9 (19.68 to 44.12)	0.66 (17%)	0.000*
Isometric hamstring strength	289.4 (±49.2)	321 (±60.5)	31.5 (19.98 to 43.20)	0.57 (11%)	0.000*
Per-Protocol					
Eccentric adduction strength	177.5 (±44.8)	213.7 (±46.5)	36.1 (21.79 to 50.58)	0.79 (20%)	0.000*
Isometric hamstring strength	288.2 (±56.5)	314.7 (±65)	26.4 (21.77 to 40.20)	0.43 (10%)	0.001*

Baseline and follow-up values are presented as mean (±standard deviation) N. * Statistically significant, $p<0.05$.

Journal Pre-proof

Journal Pre-proof

Journal Pre-proof

Figure 1. The Nordic Hamstring exercise. (A) Start position, (B) End position.

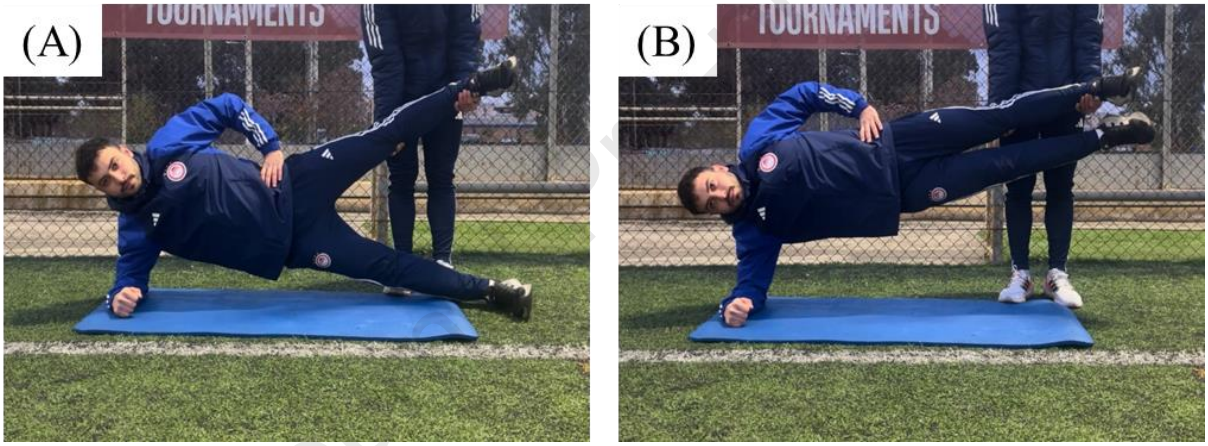


Figure 2. The Copenhagen Adduction exercise. (A) Start and end position, (B) Mid position.

Journal Pre-proof

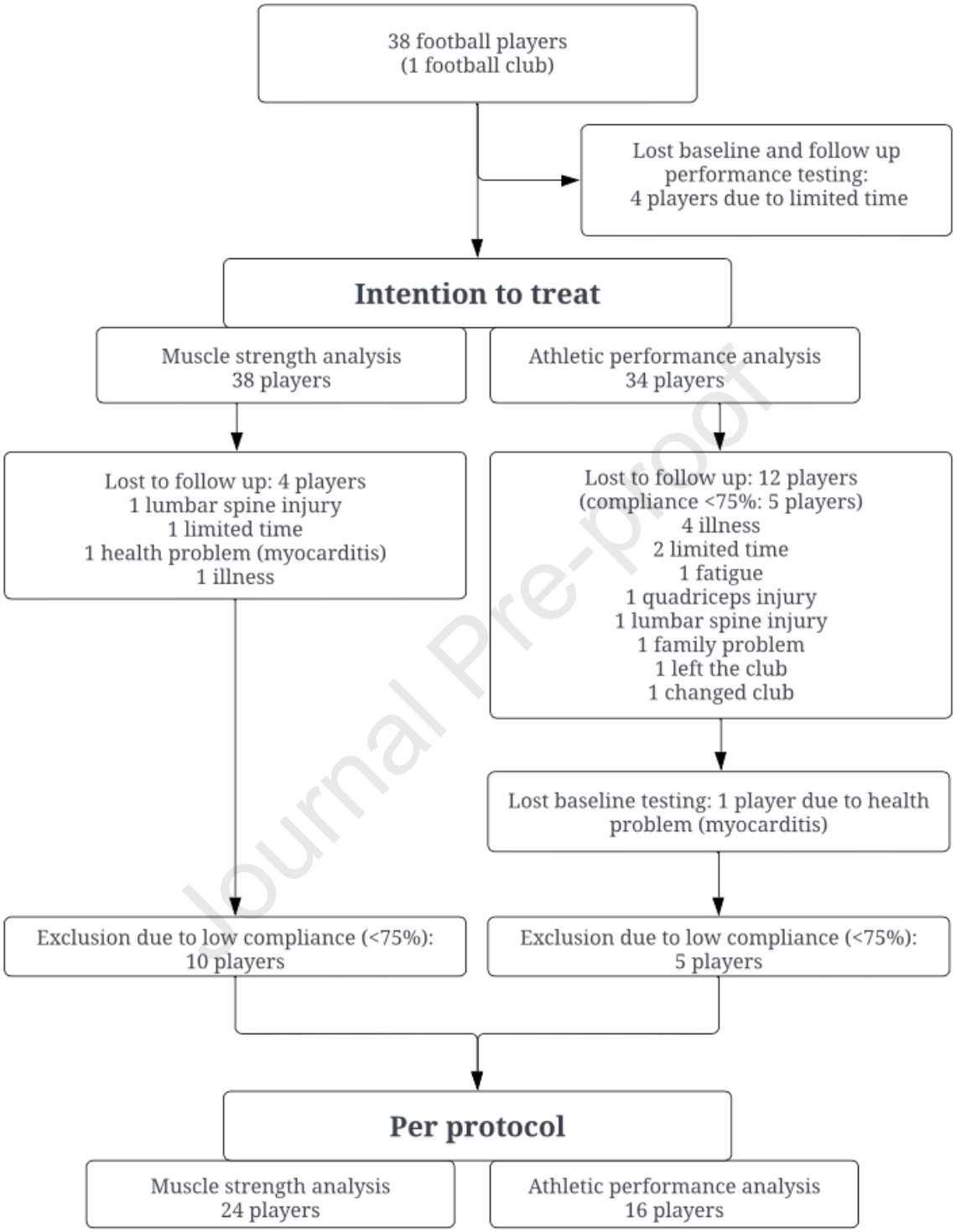


Figure 3. Flowchart of participants inclusion process

Journal Pre-proof

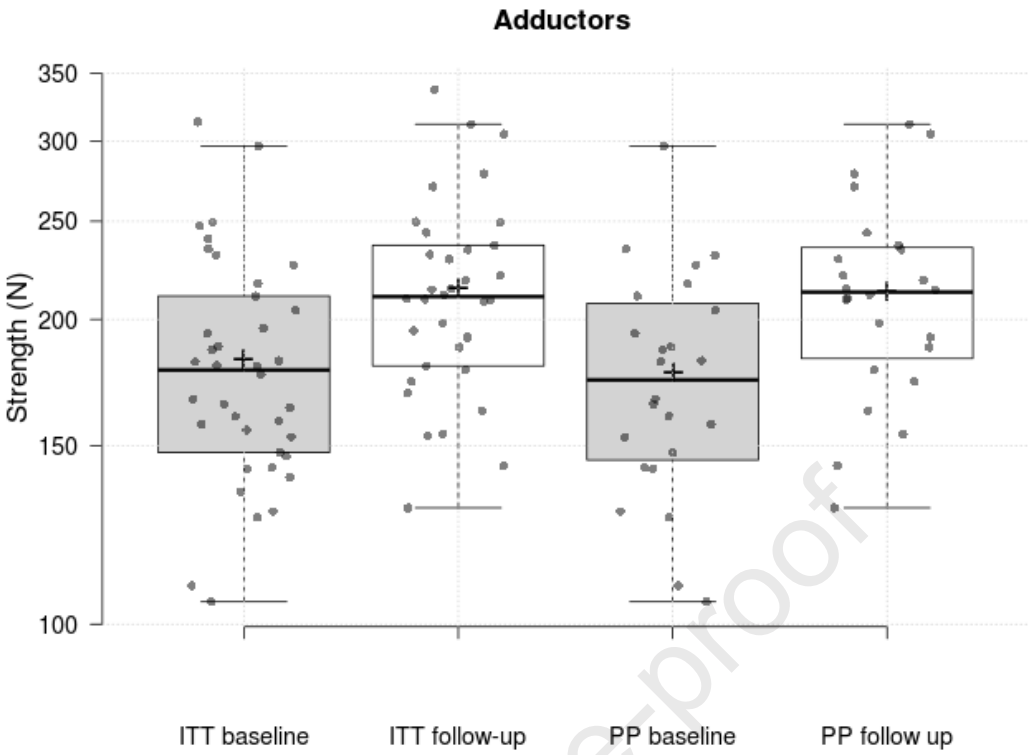


Figure 4. Eccentric hip adduction strength. Dots represent individual measurements. Plus (+) = group mean.

Journal Pre-proof

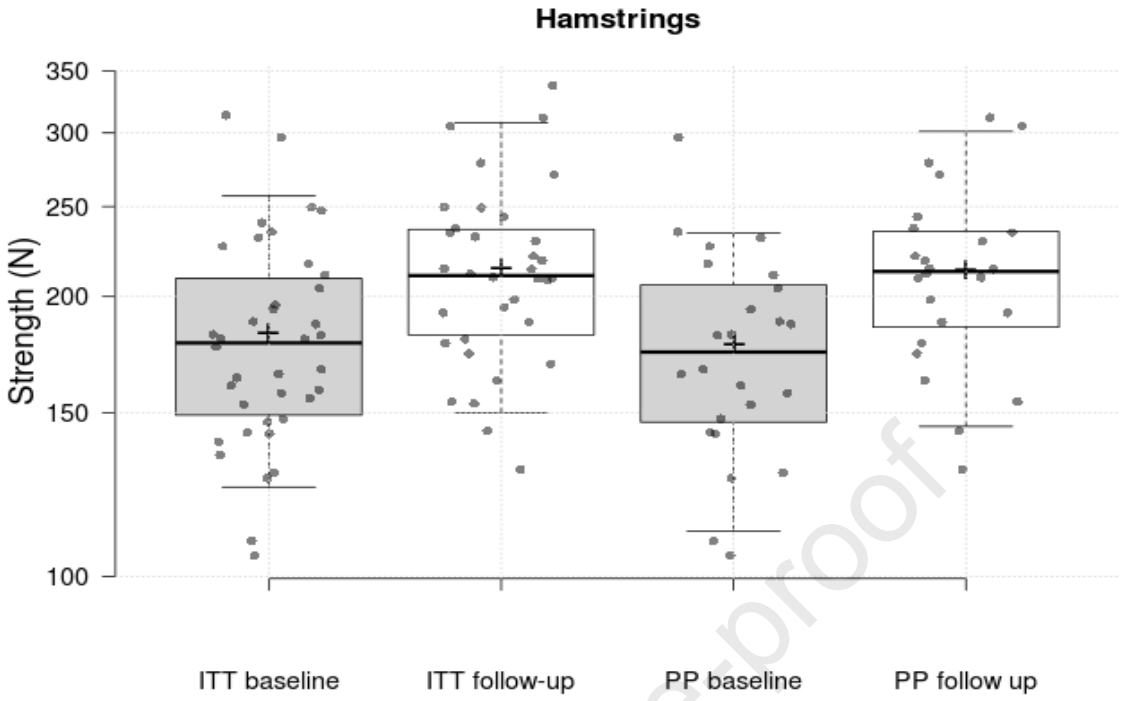


Figure 5. Isometric hamstring strength. Dots represent individual measurements. Plus (+) = group mean.

Journal Pre-proof

Journal Pre-proof

Journal Pre-proof

Ethical approval was obtained from the University of West Attica ethics committee (77055/01-09-2023).

Journal Pre-proof

Funding: This research received no external funding.

Journal Pre-proof

Journal Pre-proof

Written informed consent was obtained for every participant in accordance with the Declaration of Helsinki.

Journal Pre-proof

Journal Pre-proof

Declarations of Interest: The authors declare no conflicts of interest.

Journal Pre-proof

Journal Pre-proof

Journal Pre-proof